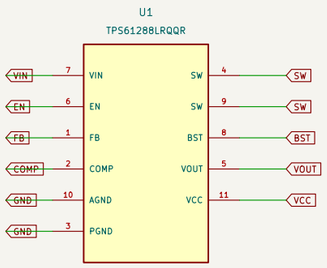
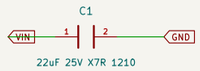
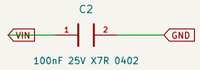
BENCH SUPPLY + ELECTRONIC LOAD ONLY. Do not connect a cell or phone.

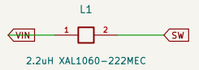
Default 5 V. JP1 closed: 9 V. Input 3–4.2 V.

25 W target at 9 V / 2.78 A. 3 A is a stress calculation, not a verified rating.

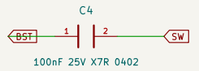




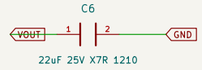


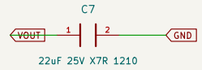




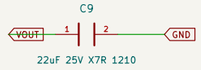


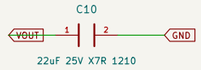












C5–C10: 132 uF nominal; effective capacitance at bias must be >=60 uF.

Feedback and loop compensation. Verify stability with load steps and Bode measurement.

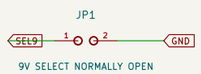






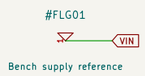










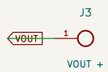


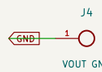


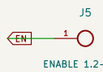
Test pads. Drive EN externally; R5 holds converter disabled when EN is open.











Matching net names connect. TI SLVSFP3C pinout and design equations. No charge circuit, UVLO or temperature cutoff.

BENCH PROTOTYPE — NO USB PD OR BATTERY PROTECTION

Sheet: /
File: spark_boost_rev_a.kicad_sch

Title: Spark boost prototype Rev A

Size: A3	Date: 2026-10-28	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1